

Computer Networks



Module 4: SNMP, ASN.1, Physical Layer

- SNMP
- ASN.1
- Physical Layer:
 - Data and signals
 - Digital transmission
 - Analog transmission
 - Bandwidth utilization
 - Transmission media

Transmission Media – Basic Concept

Transmission media are discussed in relation to the **physical layer**.

Definition

A transmission medium is anything that can carry information from a source to a destination.

Examples:

- Conversation between two people → air
- Smoke signal / semaphore → air
- Written message → mail carrier, truck or airplane

In data communications, transmission media are usually:

- Free space
- Metallic cable
- Fiber-optic cable

The information is normally a **signal** obtained by converting data from another form.

Transmission Media and the Electromagnetic Spectrum

Electromagnetic Energy

Electromagnetic energy consists of electric and magnetic fields vibrating in relation to each other.

It includes:

- Power
- Radio waves
- Infrared light
- Visible light
- Ultraviolet light
- X-rays
- Gamma rays
- Cosmic rays

Two Broad Categories

Guided Media

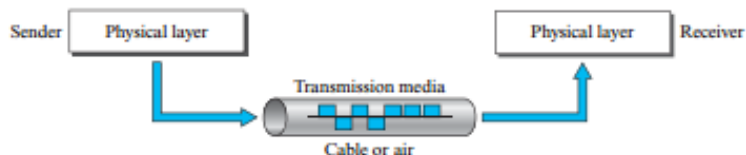
Signal travels through a physical medium.

Unguided Media

Signal travels through free space.

Figure 7.58 – Transmission Media and Physical Layer

Figure 7.58 *Transmission media and physical layer*



Guided and Unguided Media

Guided Media

Provide a physical path or conduit from one device to another.

- Twisted-pair cable
- Coaxial cable
- Fiber-optic cable

The physical boundaries of the medium direct and contain the signal.

- Twisted pair → electrical signal
- Coaxial cable → electrical signal
- Fiber optic → light signal

Unguided Medium

Free space. Electromagnetic waves travel without a physical conductor.

Twisted-Pair Cable

A twisted pair consists of:

- Two conductors, normally copper
- Each conductor has its own plastic insulation
- The two conductors are twisted together

One wire carries the signal and the other acts as a reference. The receiver uses the **difference** between the two.

Why twisting?

Noise and crosstalk can affect both wires. If the wires are parallel, the interference may affect them differently.

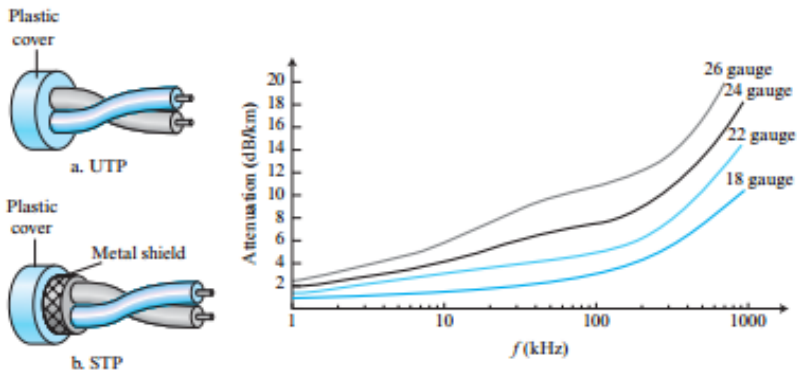
Twisting maintains a better balance, reducing the effect of unwanted signals.

Two common types:

- **UTP** – Unshielded Twisted Pair
- **STP** – Shielded Twisted Pair

Figure 7.59 – Twisted-Pair Cable and Performance

Figure 7.59 *Twisted-pair cable*



Twisted Pair – UTP, STP, Performance and Applications

UTP

- No metallic shielding
- Common in communication networks
- Less expensive and easier to handle

STP

- Metal foil or braided mesh around pairs
- Better protection from noise and crosstalk
- Bulkier and more expensive

Performance

- Can pass a wide range of frequencies
- Attenuation increases sharply above about 100 kHz
- Attenuation is measured in dB/km
- Gauge indicates wire thickness

Applications

- Telephone lines
- Local loop
- DSL
- 10Base-T
- 100Base-T LANs

Coaxial Cable

Coaxial cable carries signals at higher frequency ranges than twisted-pair cable.

Construction

- Central conductor, usually copper
- Insulating sheath
- Outer metallic conductor
- Outer insulating sheath
- Protective plastic cover

The outer metallic conductor has two functions:

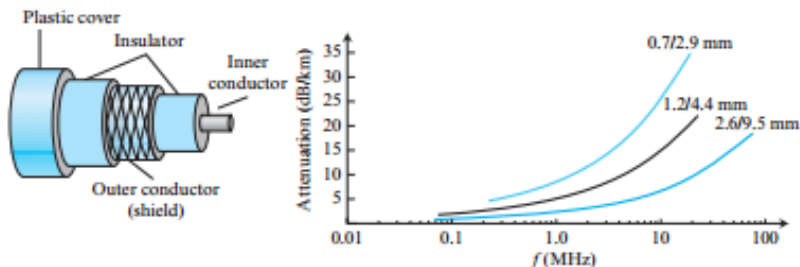
- ① Acts as a shield against noise
- ② Acts as the second conductor of the circuit

Key Point

Coaxial cable has higher bandwidth than twisted pair, but its signal attenuation is significant and repeaters may be required.

Figure 7.60 – Coaxial Cable and Performance

Figure 7.60 Coaxial cable



Coaxial Cable – Performance and Applications

Performance

- Higher bandwidth than twisted pair
- Signal weakens relatively rapidly
- Repeaters may be needed

Telephone Networks

A single coaxial network could carry about **10,000 voice signals**.

Digital telephone networks could carry data up to about **600 Mbps**.

Fiber optic has largely replaced coaxial cable in telephone networks.

Applications

- Analog telephone networks
- Digital telephone networks
- Cable TV
- Traditional Ethernet LANs

Cable TV

Modern systems commonly use hybrid fiber-coaxial networks:

- Fiber → backbone
- Coaxial → user premises

RG-59 coaxial cable is used in cable TV.

Early Ethernet LANs also used coaxial cable because of its high bandwidth and high data rate.

Fiber-Optic Cable – Basic Concept

Fiber-optic cable is made of **glass or plastic** and transmits information as **light**.

Behavior of Light

Light normally travels in a straight line through a uniform substance. When it enters another substance of different density, its direction can change.

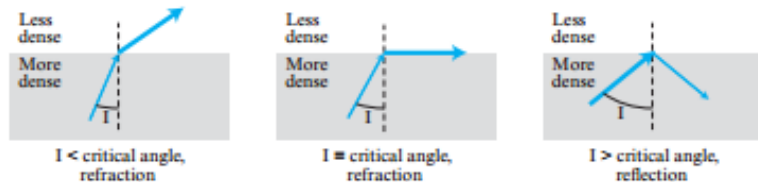
Three cases occur depending on the angle of incidence I :

- 1 $I < \text{critical angle} \rightarrow \text{refraction}$
- 2 $I = \text{critical angle} \rightarrow \text{light travels along interface}$
- 3 $I > \text{critical angle} \rightarrow \text{reflection}$

The critical angle depends on the properties of the materials.

Figure 7.61 – Refraction and Critical Angle

Figure 7.61 *Bending of light ray*



Optical Fiber – Total Internal Reflection

Optical fiber uses **reflection** to guide light through the fiber.

Construction

- Core – glass or plastic
- Cladding – less dense glass or plastic

The difference in density is chosen so that light travelling through the core is reflected at the core-cladding boundary.

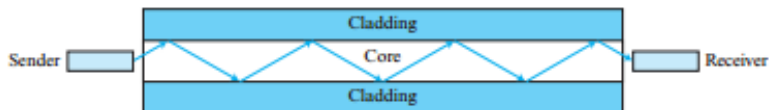
Thus:

Light signal is guided through the fiber by reflection

This allows optical fiber to transport signals over long distances with very low attenuation.

Figure 7.62 – Optical Fiber and Reflection

Figure 7.62 *Optical fiber*



Optical Fiber – Propagation Modes

Two major propagation modes are supported:

- ① **Multimode**
- ② **Single-mode**

Multimode fiber allows multiple beams to travel through the core along different paths.

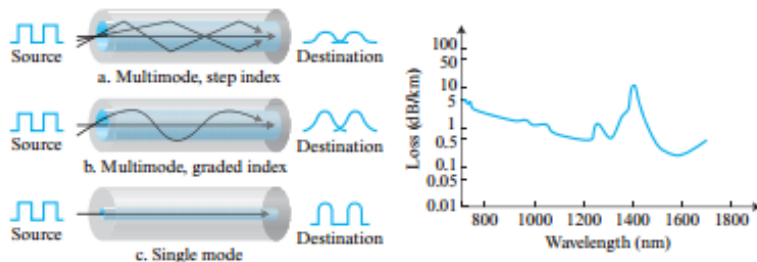
Two forms:

- Multimode step-index
- Multimode graded-index

Single-mode fiber uses a much smaller core and a highly focused light source. Beams travel almost horizontally and arrive at nearly the same time.

Figure 7.63 – Optical Fiber Propagation Modes

Figure 7.63 Modes



Optical Fiber – Step-Index, Graded-Index and Single-Mode

Type	Characteristics	Effect
Multimode Step-index	Core density remains constant. Sudden change at core-cladding boundary.	Different paths produce different delays and signal distortion.
Multimode Graded-index	Density is highest at the center and gradually decreases toward the edge.	Reduces distortion compared with step-index fiber.
Single-mode	Very small diameter, step-index fiber and highly focused light source.	Beams follow almost identical paths; delays are negligible and distortion is low.

Important

Single-mode fiber provides nearly identical propagation paths, so different beams arrive at the destination almost together.

Fiber-Optic Cable – Performance and Applications

Performance

- Attenuation is relatively flat over useful wavelengths.
- Much lower attenuation than twisted pair and coaxial cable.
- Requires fewer repeaters.
- The textbook notes approximately **10 times fewer repeaters** than the other guided media.

Applications

- Backbone networks
- SONET
- Cable TV hybrid networks
- 100Base-FX Fast Ethernet
- 1000Base-X
- 10-Gigabit Ethernet

With WDM, the text gives a data rate of approximately:

1600 Gbps

Fiber-Optic Cable – Hybrid Networks

Cable TV providers often use a combination of:

Fiber Optic + Coaxial Cable

Hybrid Network

- Optical fiber forms the high-capacity backbone.
- Coaxial cable connects the network to user premises.

Why this configuration?

The bandwidth requirement near the user is relatively narrow, so using fiber all the way to every user may not be justified.

Key idea

Fiber is used where high capacity is required; coaxial cable is used near the consumer.

Unguided Media – Wireless

Unguided media transport electromagnetic waves **without a physical conductor**.

Wireless Communication

Signals are normally broadcast through free space and may be received by any device capable of receiving them.

The wireless spectrum discussed in this section ranges approximately from:

3 kHz to 900 THz

Three major propagation methods:

- 1 Ground propagation
- 2 Sky propagation
- 3 Line-of-sight propagation

Three broad wireless groups:

Radio Waves Microwaves Infrared

Wireless Propagation Methods

1. Ground Propagation

Low-frequency radio waves travel through the lowest part of the atmosphere and follow the curvature of Earth.

Greater transmission power → greater distance.

2. Sky Propagation

Higher-frequency radio waves travel upward toward the ionosphere and are reflected back toward Earth.

Allows greater distances with lower output power.

3. Line-of-Sight Propagation

Very high-frequency signals travel directly from one antenna to another. Antennas must be directional and sufficiently tall or close together.

Table 7.1 – Electromagnetic Bands

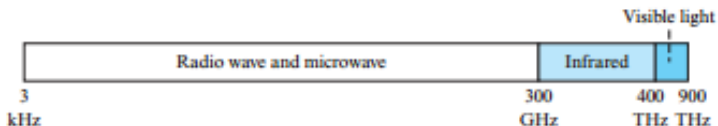
Band	Frequency	Propagation	Application
VLF	3–30 kHz	Ground	Long-range radio
LF	30–300 kHz	Ground	Radio beacons
MF	300 kHz–3 MHz	Sky	AM radio
HF	3–30 MHz	Sky	CB, ship/aircraft communication
VHF	30–300 MHz	Sky / LOS	VHF TV, FM radio
UHF	300 MHz–3 GHz	LOS	UHF TV, cellular, paging, satellite
SHF	3–30 GHz	LOS	Satellite communication
EHF	30–300 GHz	LOS	Radar, satellite

Classification

The radio and microwave region is divided into eight regulated frequency bands, from VLF to EHF.

Figure 7.64 – Wireless Electromagnetic Spectrum

Figure 7.64 *Electromagnetic spectrum for wireless communication*



Radio Waves

Radio waves are generally considered to occupy frequencies from:

3 kHz to 1 GHz

Important characteristics:

- Mostly **omnidirectional**
- Transmitting and receiving antennas do not need precise alignment
- One transmitter can reach multiple receivers
- Same-frequency transmissions can interfere with one another

Propagation and applications

- Sky propagation allows long-distance communication.
- Suitable for broadcasting.
- AM radio is a major example.
- Low and medium frequencies can penetrate walls.

Limitations

- Relatively narrow bandwidth and Lower data rates
- Most of the band is regulated by authorities

Microwaves

Microwaves occupy approximately:

1 GHz to 300 GHz

Characteristics

Microwaves are generally **unidirectional** and can be narrowly focused.

Therefore:



Transmitter antenna → Receiver antenna

must be properly aligned.

Important properties

- Line-of-sight propagation and Towers may need to be tall
- Earth curvature and obstacles can block signals
- Repeaters may be needed for long-distance links
- Very high frequencies cannot easily penetrate walls
- Wide frequency range → high possible data rate

Microwaves – Applications

Microwaves are especially useful for **unicast communication**.

Applications include:

- Cellular phones
- Satellite networks
- Wireless LANs
- Point-to-point communication

Advantages

- Directional transmission
- Less interference between properly separated antenna pairs
- Very wide bandwidth and High data rates possible

Limitations

- Requires line of sight and Obstacles can block transmission
- Long links may require repeaters
- Some frequency ranges require regulatory permission

Infrared

Infrared waves have frequencies approximately:

300 GHz to 400 THz

Corresponding wavelength range: 1 mm to 770 nm

Characteristics

- Suitable for short-range communication
- Cannot penetrate walls
- Reduces interference between rooms
- Cannot be used effectively for long-range communication

Example: Infrared remote control

The inability to penetrate walls is both an advantage and a limitation. Sunlight contains infrared radiation, which can interfere with outdoor infrared communication.

Guided Media – Comparison

Medium	Signal	Key Feature	Applications
Twisted pair	Electrical	Low cost, widely used	Telephone, DSL, LAN
Coaxial	Electrical	Higher bandwidth than twisted pair	Cable TV, Ethernet
Fiber optic	Light	Very high bandwidth, low attenuation	Backbone, SONET, Ethernet

General Trend

Twisted Pair → Coaxial → Fiber Optic

Moving toward fiber provides much greater bandwidth and lower attenuation, but requires different infrastructure and components.

Wireless Media – Comparison

Medium	Direction	Range	Examples
Radio waves	Mostly omni	Long	AM radio, broadcasting
Microwaves	Directional	Long / point-to-point	Cellular, satellite, WLAN
Infrared	Short-range	Short	Remote controls, room-scale links

Key distinction

Radio waves can generally propagate in many directions, whereas microwaves are focused and require alignment. Infrared is suitable mainly for short-range communication because it cannot penetrate walls.

Transmission Media – Key Points

Guided Media

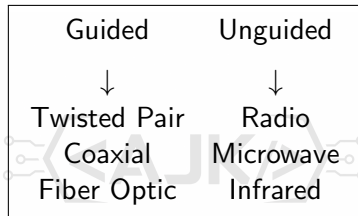
- Twisted pair – copper, inexpensive, telephone/DSL/LAN
- Coaxial – copper, higher bandwidth, cable TV/Ethernet
- Fiber optic – light, very high bandwidth, low attenuation

Unguided Media

- Radio – mostly omnidirectional, broadcasting
- Microwave – directional, line of sight, high data rate
- Infrared – short range, cannot penetrate walls

Transmission Media = Guided + Unguided

Transmission Media



Remember

- Guided media physically contain the signal.
- Fiber optic carries information using light.
- Wireless media use electromagnetic waves through free space.
- Propagation may be ground, sky or line-of-sight.